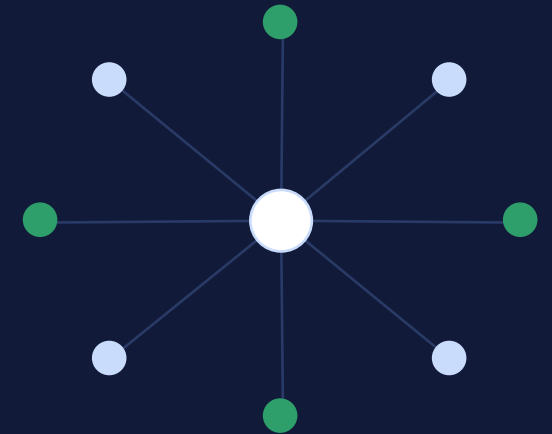


MASRA

A **M**ulti-**A**gent **S**ystem **R**esearch **A**ssistant

Eight cooperating agents · fully local models · verified, explainable reports



8

cooperating agents

147

tests passing

4

committed evidence runs

0

cloud calls



From proposal to build with feedback in mind

Unit 6 feedback flagged four real problems. Each one is answered in the build.

1

Claimed "blackboard memory", but chose CrewAI, a pipeline

Rebuilt on LangGraph: its shared state object genuinely is a blackboard.

2

"CrewAI Flows handles Ollama lazy loading" is factually wrong

Ollama itself loads and swaps models at the engine level; the code and comments now say so.

3

allow_delegation=False offered as a loop safeguard; it is not one

Replaced by three real, tested safeguards: replan cap, recursion limit, bounded model calls.

4

Blurry figures and broken references

Figures redrawn from the built architecture; references corrected.



Every deviation from the proposal is recorded with a reason in DEVIATIONS.md



Architecture: eight agents, one blackboard (Figure 1)



Nodes vs agents and Final Synthesis split

The node is the Python function, the agent is the role. Agents never talk directly: all communication is reads and writes on one shared ResearchState.



Two ranking axes, kept separate

Relevance orders sources (semantic distance to the sub-question); credibility (venue, cross-indexing, citations) only breaks ties.



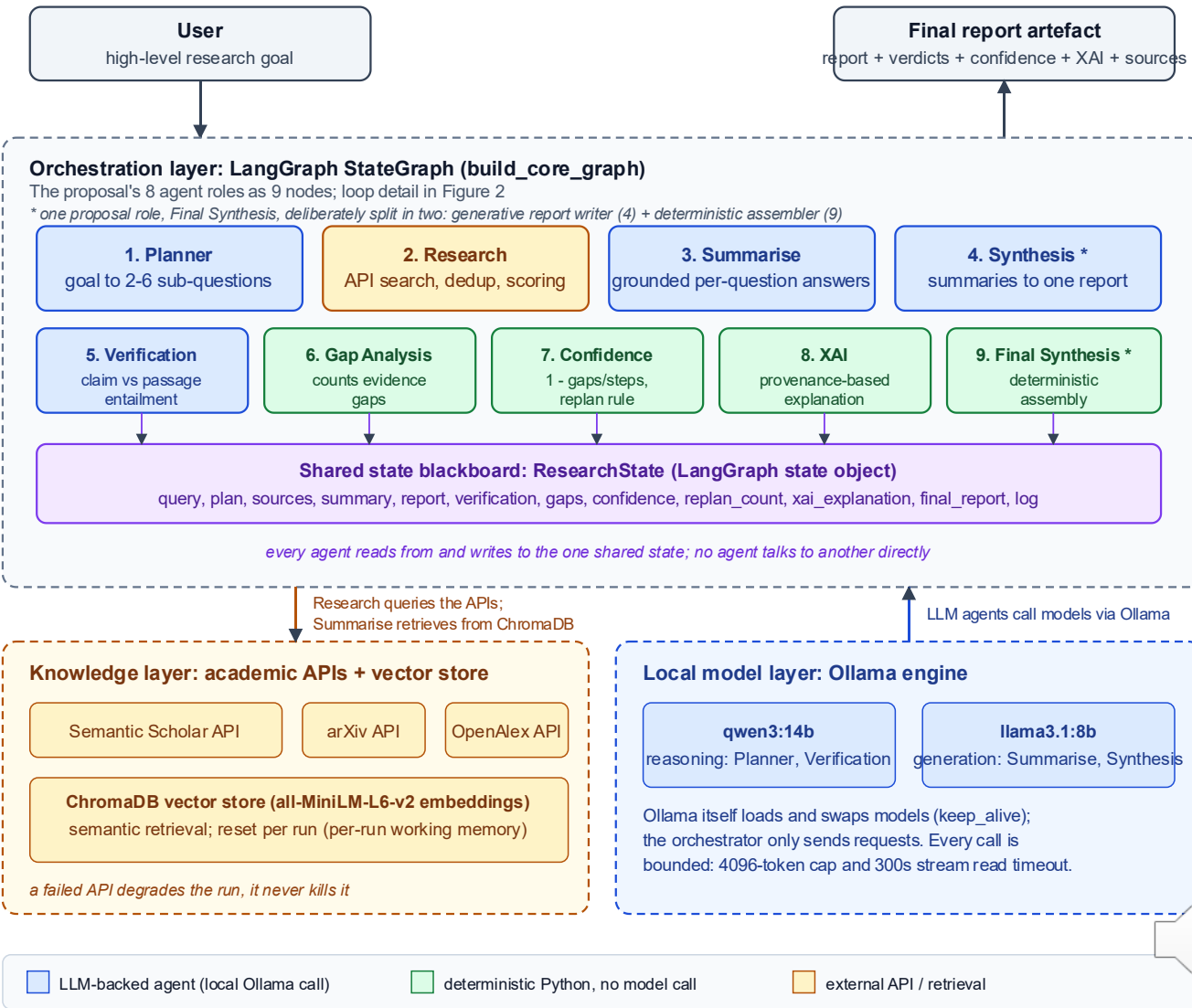
The proposal's model split, as designed

qwen3:14b for reasoning, llama3.1:8b for generation, both via local Ollama.

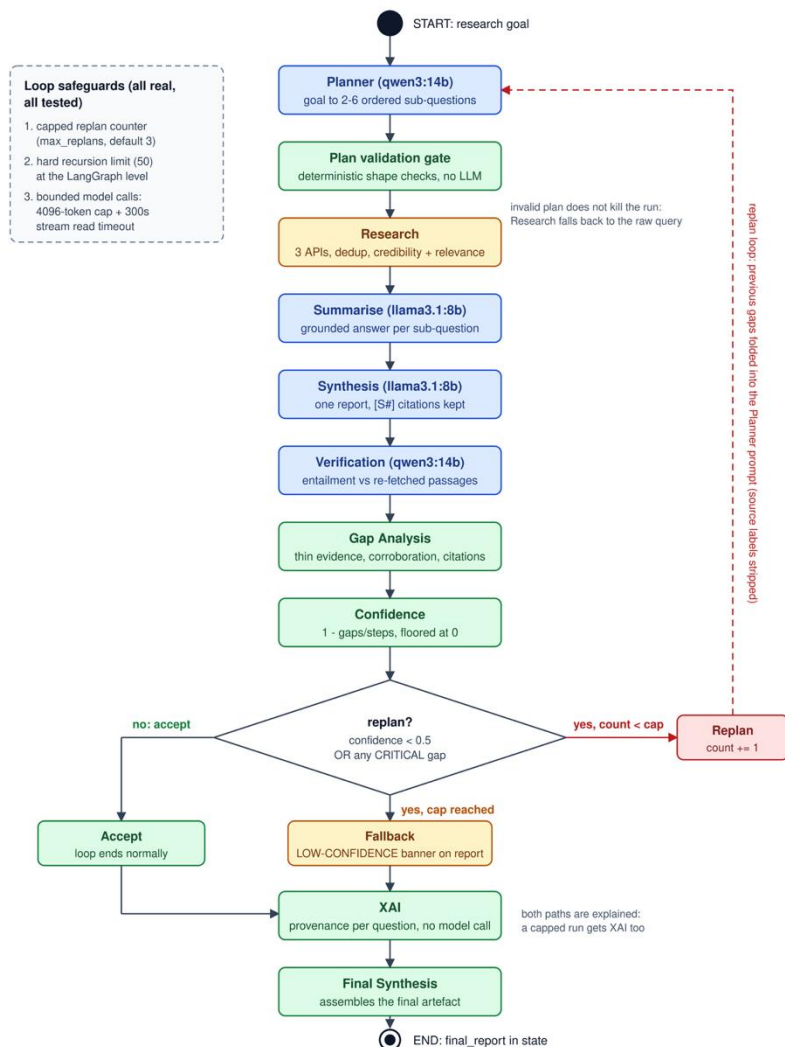


Four agents are plain Python

Deterministic, no model call: the green boxes.



The workflow and the replan loop (Figure 2)



Degrades before it dies

A malformed plan fails the deterministic shape gate; Research falls back to the raw query. A weaker run, not a dead one.



Hybrid replan rule, deliberately not one tuned number

Replan if confidence is strictly below 0.5, or if any single CRITICAL gap exists. A run at exactly 0.50 accepts.



Three bounds on the loop

Capped replan counter (3) · hard recursion limit (50) · token cap and read timeout on every model call, added after a real one-hour hang.

```
run log: python scripts/run_core.py "cheese"

confidence: 0.00 (13 gap(s) over 6 step(s))
replan: attempt 1/3 (confidence 0.00, 13 gap(s))
confidence: 0.00 (12 gap(s) over 6 step(s))
replan: attempt 2/3 (confidence 0.00, 12 gap(s))
confidence: 0.00 (12 gap(s) over 6 step(s))
replan: attempt 3/3 (confidence 0.00, 12 gap(s))
confidence: 0.00 (12 gap(s) over 6 step(s))
fallback: cap reached at 3 replan(s), confidence 0.00

LOW-CONFIDENCE REPORT - the replan cap (3) was reached without
clearing the gaps below; treat the claims here as provisional,
not verified.
```

verbatim from the committed run artefact reports/report-20260709-222223.md

The cap in action: the 'cheese' run replans three times, then hands back a flagged fallback report.



Trust: deterministic where judgment is not needed



USES A MODEL · qwen3:14b

Verification

Re-fetches the same passages each grounded answer was written from (the run's own vector store) and checks entailment: SUPPORTED / PARTIAL / UNSUPPORTED.



NO LLM · PLAIN CODE

Gap Analysis

Counts what is checkable: thin evidence, missing two-provider corroboration, UNSUPPORTED verdicts, dropped citations. Severity is CRITICAL or normal, nothing fuzzier.



NO LLM · ARITHMETIC

Confidence

1 – gaps/steps. A relative signal that steers the loop, never a probability shown to a user.

What the trust subsystem actually prints (committed runs)

```
verification: 'What are the key challenges in traditional generat...' ->  
VERDICT: SUPPORTED - Every claim in the answer is explicitly backed by the source passages
```

```
CRITICAL: 'How do advancements in cheese aging and ripening technologies...' is not  
supported by its cited sources (VERDICT: UNSUPPORTED - ...the source passages do not  
mention cheese aging/ripening technologies...)
```

```
CRITICAL: '...' - Synthesis dropped every citation (S1, S2, S3) the grounded summary  
used; the report section has no traceable source
```

```
Confidence: 0.50 | replans used: 3/3 | loop terminated by: ok
```

verbatim (elided with ...) from reports/report-20260710-110524.md and reports/report-20260709-222223.md

What the trust subsystem actually prints

Real verdicts, real CRITICAL gaps, and the accepted-run confidence line, verbatim from committed runs.



Explainability as provenance, not introspection



XAI restates recorded facts

Per question: which sources the answer used, Verification's verdict, the gaps found, and the exact confidence arithmetic.



No model self-explanation

An explanation generated by the process it claims to describe is not evidence about that process (Doshi-Velez and Kim, 2017).



Not Gap Analysis by another name

Gap Analysis diagnoses what is weak; XAI documents where every answer came from, weak or not.



The trace holds end to end

A claim in the report carries a source label like [S1]; that same label leads to the XAI block, to the appendix entry, to the live URL of the paper. The [S#] labels are run-wide and survive Synthesis, so the number never shifts. Hand-traced twice, across two providers.

One claim, traced through the real artefact

REPORT (section: primary purpose of RAG)

"The primary purpose of RAG is to mitigate hallucinations and outdated knowledge in parametric models by incorporating real-time retrieval from trusted sources [S1]."

XAI PROVENANCE BLOCK (same question, no model call)

Verification: VERDICT: PARTIAL - the primary purpose and core components are supported by the passages (the formal definition is not explicitly stated)

Sources used: [S1] A Systematic Review of Key Retrieval-Augmented Generation (RAG)

Systems: Progress, Gaps, and Future Directions, [S2] ..., [S3] ...

Gaps flagged: none

SOURCE APPENDIX ENTRY (retrieval metadata)

[S1] [dist 0.717 | score 3] (semantic_scholar) A Systematic Review of Key Retrieval-Augmented Generation (RAG) Systems: Progress, Gaps, and Future Directions - 54 citations

<https://www.semanticscholar.org/paper/2aa5f446ed125f17f0fe08827fa7a13f07ef482b>

every line verbatim from the committed run artefact reports/report-20260710-110524.md; the URL resolves to the paper

One claim traced end to end, verbatim from the flagship run.



Evidence runs on the final models

FLAGSHIP RUN

3 replans • accepted at exactly 0.50

The documented boundary, hit in the wild. Semantic Scholar 429s absorbed by paced retries; citation gaps verified genuine. report-20260710-110524.md

STRESS RUN • “CHEESE”

confidence 0.00 • cap reached • flagged fallback

No rewording can ground a vague query. The loop stops at its cap and hands back an honestly flagged low-confidence report, with clean prose in every replan. report-20260709-222223.md

CROSS-PLATFORM PAIR • WINDOWS

keyless: 46 retries, 18 losses, 3 replans → 0.50 | keyed: 3, 0, 1 → 0.60

Same query, minutes apart, with and without a Semantic Scholar key.

```
masra: scripts/run_core.py

$ python scripts/run_core.py "What is retrieval-augmented generation and why is it used?"
Running...
[ 9.2s] planner: qwen3:14b proposed 4 step(s)
[ 9.3s] validate_plan: valid=True (plan passed all checks)
[ 9.4s] research: cleared vector store for a fresh run
[ 41.7s] research: semantic_scholar 429, retrying in 30s (attempt 1/2)
[ 93.2s] research: 21 raw results -> 18 after dedup
[ 94.6s] research: embedded 18 abstracts into ChromaDB
[ 128.9s] summarise: grounded 'What is the definition of retrieval-augmented gene' on 3 passages
[ 193.4s] synthesis: llama3.1:8b wrote a 4845-char report
[ 246.0s] verification: 'What is the definition of retrieval-augmented gene' -> VERDICT: SUPPORTED ...
[ 271.8s] gap_analysis: grounding overlap: 3 distinct paper(s) across 12 grounding slot(s)
[ 271.9s] confidence: 0.00 (4 gap(s) over 4 step(s))
[ 271.9s] replan: attempt 1/3 (confidence 0.00, 4 gap(s))
... two more replan rounds, previous gaps folded into the Planner prompt ...
[1571.2s] gap_analysis: grounding overlap: 5 distinct paper(s) across 18 grounding slot(s)
[1571.3s] confidence: 0.50 (3 gap(s) over 6 step(s))
[1571.3s] accept: confidence 0.50, loop ends normally
[1571.4s] xai: explained 6 question(s) from recorded provenance
[1572.0s] final_synthesis: assembled a 18508-char final report

Saved to reports/report-20260710-110524.md
```

Live per-node progress from a real run.



One honest note: no live run has produced the critical override alone at high confidence. That arm's evidence is its two unit tests, by construction.



What manual testing found (and the tests that pin it)

15 documented defects 15 regression tests 147 tests passing

- **The corroboration check almost never fired**
Looked correct, passed its tests; real output showed the definition was wrong. Caught by reading a report critically.
- **The citation check was broken from two sides**
Two different models each violated a format assumption the other respected. Dev-model runs proved nothing about the final model.
- **Replan prompts leaked [S#] labels into live API queries**
Fixed with a deterministic strip, not a prompt instruction: a strip is checkable, a prompt is a hope.
- **One unbounded model call hung a run for an hour**
Every model call is now bounded by a token cap and a read timeout.
- **A Windows run exposed an encoding defect**
One no macOS run could ever show. One OS proves nothing about another.

```
test suite output (committed captures)

$ pytest -q
.....SS..... [ 48%]
..... [ 96%]
147 passed, 2 skipped in 3.49s

$ MASRA_FUNCTIONAL=1 pytest tests/test_core_graph.py -s
tests/test_core_graph.py (full graph, live Ollama + APIs, both paths)
2 passed in 1554.08s (0:25:54)

verbatim from evidence/pytest-output.txt and evidence/pytest-functional-output.txt
```

“A month of dev-model runs proved nothing about the final model's habits; one operating system proves nothing about another.”

Honest limitations and deviations

The project's premise is not claiming behaviour the system does not have. Limitations are stated as plainly as features.



Per-run memory, by design

The vector store resets each run: In future, we could keep a persistent vector store of papers tagged by run, instead of clearing it each time. .



Grounding correlation on small corpora

Similar sub-questions can ground on the same top papers, so per-question checks are correlated. The overlap is logged on every run and documented, not engineered away.



Section-level traceability, not per-claim markers

The citation check guarantees each section stays traceable to its source set. Claim-level alignment would need exactly the fuzzy judgment the deterministic checks exist to avoid.



Descoped from the proposal

PubMed, Crossref, SerpAPI and the web UI: each descoped item is recorded with its reason in DEVIATIONS.md.

Answered in code, not words

“

Trustworthiness came less from adding intelligence and more from removing it where it was not needed.

deterministic checks · arithmetic confidence · provenance explanations ·
a human reading real output critically

KEY REFERENCES (HARVARD)

Doshi-Velez, F. and Kim, B. (2017) · Lewis, P. et al. (2020) · Rao, A.S. and Georgeff, M.P. (1995) · Wooldridge, M. (2009) · Yao, S. et al. (2023) · full list in README.md

Thank you

